

Total Transport for the Cluster-Quasimodel Transition System: Round-18, Implementing M1–M6 — with the Birth Flip

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Abstract

The twelfth review rejected round 17 at a precise seam: the ordered-pair transport rule covered only pairs whose endpoints are both updated by the ambient transition, omitting mixed pairs with a parent-copy endpoint, so the pair condition could miss operational pairs. This delta implements the review’s repair package with one sharpening that working out the path semantics makes unavoidable: the total endpoint update U has *three* clauses, not two. Besides the review’s ambient clause (recorded steering rows carried forward) and resident clause (catalogue read-off), cross-branch occurrences require a *birth-flip* clause: a younger occurrence’s entry to an older separator member is the converse of the steering value that the *older* side received at the younger’s birth. With U total, pairs are transported in lockstep, the inclusion theorem is re-proved by a five-way endpoint-class analysis, and an order-invariance lemma gives the least-common-ancestor path semantics the review asked for. The review’s syntactic demands are also implemented: injective slot maps (whose necessity the new harness independently re-derived when non-injective maps produced conflicting pattern writes), the continued/core consistency invariant, and a sibling-ordering convention completing the witness map. The harness architecture answers Finding 17.3 structurally: rule-computed states, derived without ever reading the realized frame, are cross-checked against frame truth on genuinely branching unfoldings — 150 runs, zero divergences, 2,820 mixed pairs and 140 birth-flip entries exercised, a constructed negative control, and hash-seed-deterministic output. Open and unchanged: the width budgets (F6) and uniformization (W2’). Status: *strongly supported, not certified*.

Contents

1	Scope, and one sharpening of the work order	2
2	The total endpoint update	2
3	Order invariance and the path semantics (M5)	3
4	The inclusion theorem, re-proved (M4)	4
5	Certificate syntax (17.4), the overlap invariant (17.5), and the sibling convention (R6)	4

*Round-18 delta on rounds 15–17, implementing the twelfth review’s repair package M1–M6 (papers/gpt5.5.round17_review/). Everything not amended is inherited. Ledger going in: twelve reviews — eleven defects with witnesses, one no-counterexample verdict. This delta is unreviewed; presume it contains the next finding. Designated weak points: Section 8.

6	Machine checks (WP32)	5
7	Response map	5
8	What would invalidate this delta	5
9	Status	6

1 Scope, and one sharpening of the work order

The twelfth review’s M1 proposed a total endpoint update with two clauses: ambient endpoints keep their T -images; parent-copy endpoints take birth-adjacent catalogue states. Implementing M5 (the path semantics for “later gluings elsewhere”) shows a third case is forced. Consider an occurrence y born in a copy A attached *after* some copy k elsewhere in the tree, and a later gluing h whose separator contains a member $s \in V_k$. The value $\rho(y, s)$ was not assigned when k was attached (y did not exist) nor by y ’s own steering at any later step: it was assigned *at y ’s birth*, when s was a pre-existing occurrence steered at A and received the value $f_A(\text{St}_A(s))(y)$ toward the fresh y . The entry $y \rightarrow s$ is therefore the *converse of the older side’s steering value at the younger’s birth*. We call this the *birth flip*. It is the least-common-ancestor role change that the twelfth review’s Finding 17.2 said was undefined; making it a named clause of U is the content of this delta beyond the literal work order.

2 The total endpoint update

Fix a certificate in the normal form of round 17 (Q3’), tightened by Section 5 below. For occurrences x, s with s a member of the separator σ_h of a gluing h (so s is a member of h ’s parent copy k , or a core reference), the entry $\rho(x, s)$ is given by exactly one of:

Definition 2.1 (U: three clauses). Let $b(x)$ denote the attachment step of x ’s birth copy, with core occurrences born at the base.

(U-core)

$s \in \text{core}$: the persistent core-row entry of x (in-pattern at x ’s birth, by (N3)).

(U-res)

$x \in V_k$ (a co-member of s ’s copy): the pattern value $N_k(x, s)$, a catalogue value.

(U-down)

$b(x) < b(s)$ and $x \notin V_k$: x was steered when k was attached; the entry is the recorded steering value $f_{g_k}(\text{St}_{g_k}(x))(s)$.

(U-flip)

$b(x) > b(s)$: at x ’s birth step, every pre-existing occurrence is either a member of x ’s birth copy or steered there (there is no third possibility — Lemma 2.2); the entry is, respectively, the converse of the in-pattern value $N_A(\cdot, x)$, or the converse of the recorded steering value $f_{g_A}(\text{St}_{g_A}(s))(x)$.

$U_{* \rightarrow h}(x)$ is the state of x at h : its type, persistent core row, the σ_h -row assembled from the clauses above, and its certificate-determined safety signature.

Lemma 2.2 (Birth dichotomy). *At every attachment step, every pre-existing occurrence is a member of the new copy (equivalently, of its separator, by (N1)) or is steered at that step. Hence every pair of occurrences receives its value exactly once — in-pattern if ever co-patterned, else by steering at the younger’s birth — and (U-down)/(U-flip) always find a recorded value.*

Proof. The construction steers precisely the pre-existing occurrences outside the new copy; (N1) says the pre-existing members of the new copy are exactly the separator. Single assignment is round-17’s no-overwrite lemma, unchanged. The recorded value exists because the younger’s birth step steered (or co-patterned) the older endpoint, by the first sentence. $\square$

Lemma 2.3 (U is total, finite, and catalogue-computable (M1/M3)). *The four clauses are exhaustive and mutually exclusive on the pairs (x, s) with $s \in \sigma_h$, and each reads finitely many values from the catalogue, the certificate’s steering functions, and states already computed at earlier steps. The state alphabet is unchanged from round 17.*

Proof. Exhaustiveness: $s \in \text{core}$ or $s \in V_k \setminus \text{core}$; in the latter case $x \in V_k$ or not, and if not, $b(x) \neq b(s)$ (distinct copies) splits into (U-down)/(U-flip); Lemma 2.2 supplies the recorded value in both. Computability: (U-core), (U-res) are catalogue lookups; (U-down), (U-flip) are lookups into steering values $f(\varsigma)(\cdot)$ at states ς computed at strictly earlier steps — well-founded in the attachment order. $\square$

Definition 2.4 (Pair transport under U (M2)). Replace round-17’s (P3) by: if $(\varsigma, \varsigma', W)$ is reachable before a step and both endpoints persist, then $(U(\varsigma), U(\varsigma'), W)$ is reachable after it, together with its converse. Creation (P2) is retained *in both orientations*: at a birth step, for every steered or co-patterned pre-existing x and fresh y , both $(\text{St}(x), \text{St}(y), V)$ and $(\text{St}(y), \text{St}(x), V^{-1})$ enter the pair set, where V is the assigned or in-pattern value. Co-patterned seeds (P1) are unchanged.

3 Order invariance and the path semantics (M5)

Lemma 3.1 (Order invariance). *The value of every pair, and hence every state $U_{* \rightarrow h}(x)$, depends only on the attachment tree (which copies attach where, with which slot maps) and not on the interleaving order in which incomparable branches are expanded.*

Proof. By Lemma 2.2, the value of a pair (x, y) is fixed at the later of the two birth steps, by data local to that step: the younger’s birth copy, slot map, and the older’s state there. The older’s state at that step is, inductively, a function of pair values fixed at even earlier birth steps along the relevant path. No step touches a pair whose endpoints both pre-exist (no overwrite), so expanding an incomparable branch neither changes existing values nor the states read at later births. Induction on the tree. $\square$

Corollary 3.2 (LCA path semantics). *For occurrences y, z and a gluing h , the pair state $(\text{St}_h(y), \text{St}_h(z), \rho(y, z))$ is determined by the attachment-tree paths from the birth copies of y and z through their least common ancestors to h ’s parent, with (U-down) applying along descents on the elder side and (U-flip) applying at and above the younger’s birth. “Later gluing elsewhere in the tree order” (round-17’s undefined phrase) is thereby replaced: the fixed point closes under one-step attachment, and Lemma 3.1 makes the result path-functional.*

4 The inclusion theorem, re-proved (M4)

Theorem 4.1 (Inclusion, total-transport form). *In every finite unfolding prefix of a certificate in normal form: at every gluing, every existing occurrence’s state and every ordered pair $(\text{St}(y), \text{St}(z), \rho(y, z))$ lie in the fixed point computed with seeds (P1), two-sided creation (P2), and U-transport (Definition 2.4).*

Proof. Induction over attachment steps, as in round 17, with case (c) replaced by a five-way endpoint-class analysis. At a step, classify each pre-existing endpoint as *resident* (member of the new copy, i.e. separator or core) or *steered*; steered endpoints subdivide by Definition 2.1 into down and flip histories, but their state update is uniformly U. For a pair of pre-existing occurrences, the five combinations are: *steered/steered* — transported by Definition 2.4, both endpoints U-updated, W unchanged; *steered/resident* and *resident/steered* — transported likewise: U is total, so the resident endpoint’s update (U-res) is a legitimate transport step (this is exactly the mixed case round 17 omitted and Finding 17.1 witnessed); *resident/resident* — co-patterned at this step, hence a (P1) seed, whose value agrees with any pre-existing value by gluing coherence and no-overwrite. Pairs with a fresh endpoint are (P2) creations, in both orientations. Fresh/fresh pairs are (P1) seeds. States: residents take catalogue-computable states; steered occurrences take U-images of inductively-reachable states. Order invariance (Lemma 3.1) licenses treating “the next gluing” in tree order without loss of generality. $\square$

Corollary 4.2. *Round-16’s Theorem 3.2 (deterministic canonical completion) holds with (S4) checked on the fixed point of Definition 2.4: every operational ordered pair — including the mixed ambient/resident and cross-branch flip pairs — is present with its current states and realized mutual value. The twelfth review’s witness geometry is rejected: the pair $(\text{St}_h(y), \text{St}_h(z), \text{PP})$ is transported, S4 demands $f(\text{St}_h(y))(b) \in \text{Comp}(\text{PP}, f(\text{St}_h(z))(b))$, and the bad (PO, DR) choice violates $\text{Comp}(\text{PP}, \text{DR}) = \{\text{DR}\}$ (WP32 part B2).*

5 Certificate syntax (17.4), the overlap invariant (17.5), and the sibling convention (R6)

Definition 5.1 (Port discipline). Every template’s ports are partitioned into core ports, inherited slots, and fresh ports. Every gluing declaration maps the child’s inherited slots *injectively* into the parent’s non-core ports plus core; fresh occurrence identities are allocated per copy; non-core identity reuse occurs only through slot maps (connected traces). (N1) of round 17 is thereby a syntactic consequence of the declarations, not a run property.

Remark 5.2 (Injectivity is not decorative). WP32’s own development re-derived its necessity: with a non-injective slot map, one occurrence occupies two pattern ports, the instantiation performs conflicting writes, and the rule-computed states diverge from frame truth — the harness caught exactly this before injectivity was enforced, after which agreement is exact (150 runs, 0 divergences). This is the twelfth review’s 17.4, confirmed empirically.

Remark 5.3 (Continued/core overlap (17.5)). Core members are both persistent core-row indices and separator members at every step. The two sources agree by construction: (U-core) and (U-res)/(U-down)/(U-flip) coincide on core columns because core values are in-pattern at every copy (N3) and never overwritten. We state this as an explicit invariant so a mechanization can discharge it once.

Remark 5.4 (Sibling ordering (R6 completion)). Children of a copy are ordered lexicographically by (source member index, demand index in closure order); demands designating the same gluing edge at the same source share one copy. With this convention and the witness map of round 17, the unfolding is a function of the certificate, including its tree structure.

6 Machine checks (WP32)

`verification/python/wp32_round18_total_transport.py`, all PASS, with the harness architecture the twelfth review’s Finding 17.3 demanded: rule-computed states (never reading the frame) cross-checked against frame truth on genuinely *branching* unfoldings. **A** (M1/M5): 150 random branching unfoldings, *zero* U-clause divergences between rule and frame. **B** (M2/17.1 regression): 2,820 mixed ambient/resident and flip pairs exercised, 140 birth-flip entries; the WP31 geometry’s mixed pair is transported and S4 rejects the bad row. **C** (M6): a constructed adversarial steering violating S4 produces closure failures in 17/41 runs — the negative control fires (round-17’s harness never sampled one). **D** (17.3): no set-iteration choice points; a result digest is printed and is stable across PYTHONHASHSEED values.

7 Response map

M1	U total on all pre-existing endpoints (Definition 2.1), with the birth-flip clause added beyond the work order’s two cases (Section 1).
M2	Pair transport under U with converses; two-sided creation (Definition 2.4).
M3	Lemma 2.3.
M4	Theorem 4.1: five-way endpoint-class analysis replacing “both endpoints update by T ”.
M5	Lemma 3.1 and Corollary 3.2: one-step closure plus order invariance; the LCA characterization derived, the undefined phrase eliminated.
M6	WP32: branching trees, mixed transport exercised and counted, constructed negative S4 control, deterministic output.
17.4	Definition 5.1 (with empirical confirmation).
17.5	The overlap invariant remark.
R6	The sibling convention remark.

8 What would invalidate this delta

- (Z1) **A gap in the birth dichotomy or the flip clause:** a relative position of (x, s) not covered by Definition 2.1 — e.g. if some construction step neither co-patterns nor steers a pre-existing occurrence (this would contradict Lemma 2.2’s first sentence, which is definitional here, but an extraction-side presentation might demand such a step).
- (Z2) **Order invariance under declarations richer than the normal form:** Lemma 3.1 leans on no-overwrite and injective slot maps; exotic-but-legal certificates are the place to look.
- (Z3) **The standing open items:** F6 (width budgets), W2’ (uniformization), and the single external input (RCC5 patchwork).

9 Status

Twelve reviews: eleven defects with witnesses, one no-counterexample verdict. This delta implements the twelfth review’s full repair package, sharpened by the birth-flip clause that the path semantics forces, and its harness now checks the *text* against frame truth by construction — the mismatch class that produced Findings 17.1 and 17.3 is structurally excluded from the toolchain, not just from this round. The delta is unreviewed; presume the next finding. The soundness half remains, as of this writing, fully written and machine-cross-checked; the completeness half retains its two named open lemmas. The decidability theorem for $\mathcal{ALCI}_{\text{RCC5}}$ remains **strongly supported, not certified**.

References

- [1] GPT-5.5 (OpenAI). *Formal Response on the Round-17 Transition-System Delta*, July 13, 2026. [papers/gpt5.5_round17_review/](#).
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- [4] J. Renz and B. Nebel. On the complexity of qualitative spatial reasoning. *Artificial Intelligence* 108 (1999).